

Database System Principle - Introduction

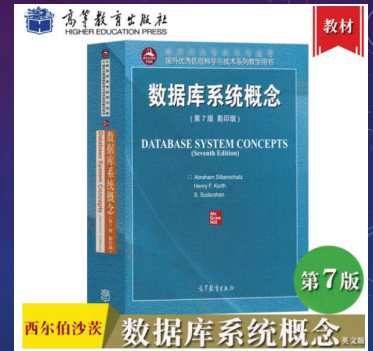
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OBJECTIVES

- Database system
- Data Abstraction
- Database Model
- Database Languages
- Relational Database
- Database Design
- Database Architecture
- History of Database

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DATABASE SYSTEM

- Database-management system
 - DBMS
 - A collection of interrelated **data** and a set of **programs** to access those data
- Goal of DBMS
 - Provide a way to **store** and **retrieve** database information that is both **convenient** and **efficient**
 - **security**

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DATABASE-SYSTEM APPLICATIONS

- Enterprise Information
 - Sales, accounting, human resources, CRM, ERP, manufacturing, online retailers
- Banking and Finance
 - Banking, credit card transactions, finance stock
- Universities: Students, courses, grades
- Airlines : reservations, schedule information
- Telecommunication: records of calls made, bills
- ...

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PURPOSE OF DATABASE SYSTEMS

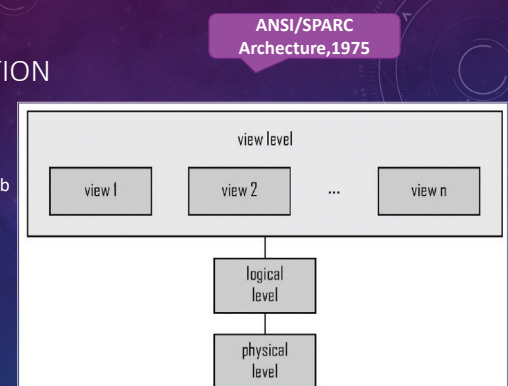
- Enable durability (持久性) storage
- Large amounts of data
- Isolation (独立性) among users
- Atomicity(原子性)
- Reduce redundancy(冗余)
- Consistency(一致性)
- Integrity (完整性)
- Efficient store and access
- Concurrent(并发) access
- Security(安全)

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DATA ABSTRACTION

- View level
 - 视图层:外模式
 - External view: part of db
- Logical level
 - 逻辑层:模式
 - Conceptual view: what
- Physical level
 - 物理层:内模式
 - Internal view: how

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EXAMPLE OF DATA ABSTRACTION: LOGICAL VIEW 概念模式

```
type instructor = record
  ID : string;
  name : string;
  deptname : string;
  salary : integer;
  birth : date;
  gender : boolean;
  regDate : datetime;
end;

type department = record
  ...
end;

type student = record
  ...
end;
```

EXAMPLE OF DATA ABSTRACTION: EXTERNAL VIEW 外模式

```
type instructor_computerdepart = record
  ID : string;
  name : string;
  salary : integer;
  birth : date;
  gender : boolean;
  regDate : datetime;
end;

create view instructor_computerdepart
as
select
  ID,name,salary,birth,gender,regDate
from instructor
where deptname='Computer'
```

EXAMPLE OF DATA ABSTRACTION: PHYSICAL VIEW 内模式

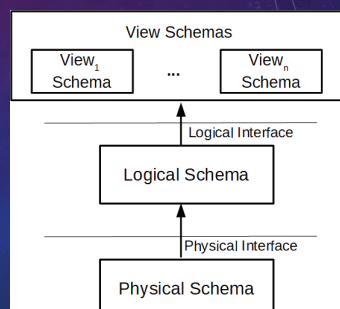
```
create table instructor
  (ID      varchar(5) PRIMARY KEY,
   name    varchar(20) not null,
   ...
   regDate datetime not null,
  )ENGINE=InnoDB
PARTITION BY RANGE(YEAR(regDate)) (
  PARTITION p_1949 VALUES LESS THAN (1949),
  PARTITION p_1979 VALUES LESS THAN (1979),
  PARTITION p_curr VALUES LESS THAN MAXVALUE);
```

DATABASE INSTANCES AND SCHEMAS

- DB Instance 实例
 - The collection of information stored in the database at a particular moment
- DB Schema 模式
 - The overall design of the database
 - Physical schema,
 - logical schema,
 - Subschema (view level)

DATABASE INSTANCES AND SCHEMAS

- Logical data independence
 - (逻辑数据独立性)
- Physical data independence
 - (物理数据独立性)



DATA MODEL

- A collection of tools for describing
 - Data
 - Data relationships
 - Data constraints
 - Data semantics

DATA MODEL

• Alternative View of Data Model

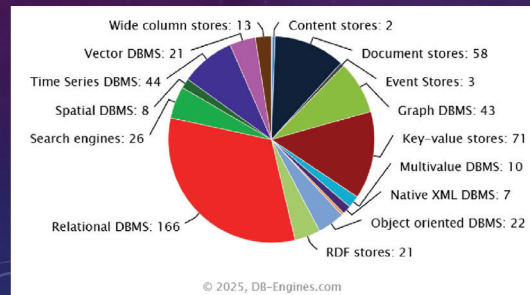
- Data structure 数据结构
 - 静态特性
- Data Operation 数据操作
 - 动态特性
- Integrity Constraint 完整性约束

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DATA (DATABASE) MODELS

<https://db-engines.com/en/ranking>

https://db-engines.com/en/ranking_categories



DB-Engines lists 424 different database management systems, which are classified according to their database model (e.g. relational DBMS, key-value stores etc.).

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DATA (DATABASE) MODELS

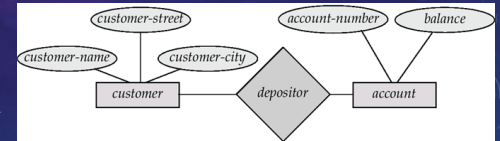
- Entity-relationship model(实体关系E-R)
- Hierarchical data model(层次)
- Network data model(网络)
- Relational data model(关系)
- Object-based data model(对象)
- Semi-structured data model (半结构)
- Graph data model (图)

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ENTITY-RELATIONSHIP MODEL

• Entity-relationship model

- 实体关系模型E-R
- Entity
- Relationship
- Attribute 属性



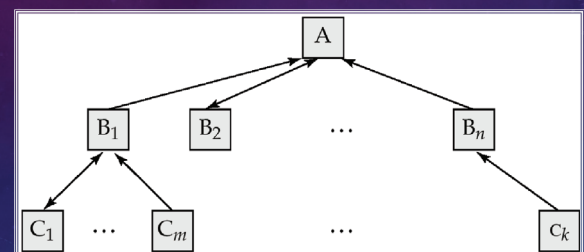
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HIERARCHICAL DATA MODEL

- A hierarchical database consists of a **collection** of **records** which are connected to one another through **links**.
- a **record** is a collection of fields, each of which contains only one data value.
- A **link** is an association between precisely two records.
- The hierarchical model differs from the network model in that the records are organized as collections of trees rather than as arbitrary graphs.

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HIERARCHICAL DATA MODEL



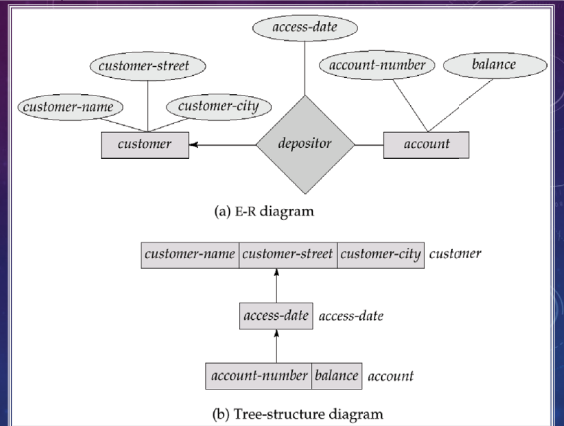
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HIERARCHICAL DATA MODEL

- The schema for a hierarchical database consists of
 - boxes**, which correspond to record types
 - lines**, which correspond to links
- Record types are organized in the form of a *rooted tree*.
 - No cycles** in the underlying graph.
 - Relationships formed in the graph must be such that only **one-to-many** or **one-to-one** relationships exist between a parent and a child.

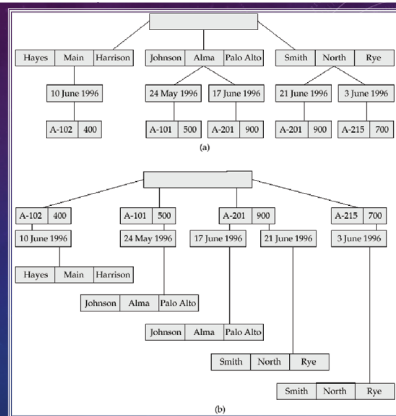
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HIERARCHICAL DATA MODEL: CASE



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HIERARCHICAL DATA MODEL: CASE



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NETWORK DATA MODEL

- Data are represented by collections of *records*.
 - similar to an entity in the E-R model
- Records and their fields are represented as *record type*
- Relationships among data are represented by *links*
 - similar to a restricted (binary) form of an E-R relationship
 - restrictions on links depend on whether the relationship is **many-many**, many-to-one, or one-to-one.

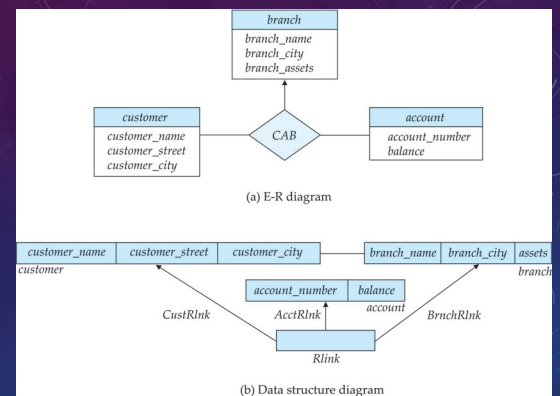
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NETWORK DATA MODEL

- Schema representing the design of a network database.
- A data-structure diagram consists of two basic components:
 - Boxes**, which correspond to record types.
 - Lines**, which correspond to links.
- Specifies the overall logical structure of the database.

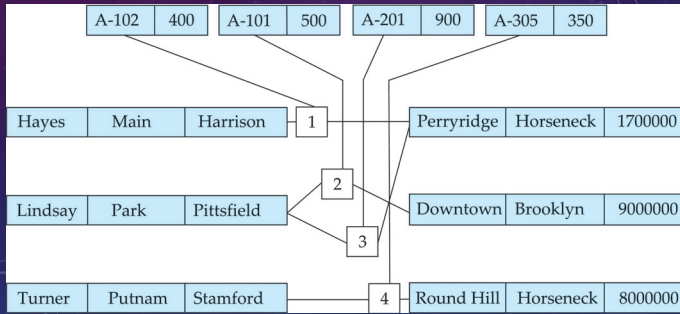
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NETWORK DATA MODEL : CASE

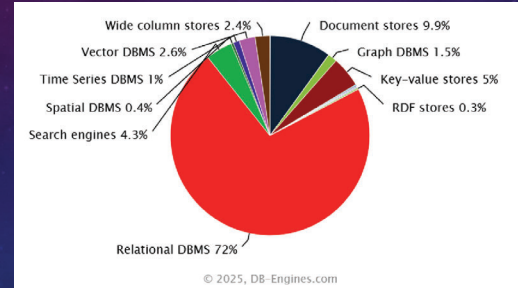


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NETWORK DATA MODEL: CASE



RANKING SCORES PER DB CATEGORY IN PERCENT, FEBRUARY 2025



RELATIONAL DATA MODEL

- Relational data model
 - A collection of tables
 - Table
 - A collection of records
 - Multiple columns
 - Relation

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RELATIONAL DATA MODEL: CASE

customer_id	customer_name	customer_street	customer_city	account_number
192-83-7465	Johnson	12 Alma St.	Palo Alto	A-101
192-83-7465	Johnson	12 Alma St.	Palo Alto	A-201
677-89-9011	Hayes	3 Main St.	Harrison	A-102
182-73-6091	Turner	123 Putnam St.	Stamford	A-305
321-12-3123	Jones	100 Main St.	Harrison	A-217
336-66-9999	Lindsay	175 Park Ave.	Pittsfield	A-222
019-28-3746	Smith	72 North St.	Rye	A-201

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RELATIONAL DATA MODEL: CASE

customer-id	customer-name	customer-street	customer-city
192-83-7465	Johnson	12 Alma St.	Palo Alto
019-28-3746	Smith	4 North St.	Rye
677-89-9011	Hayes	3 Main St.	Harrison
182-73-6091	Turner	123 Putnam Ave.	Stamford
321-12-3123	Jones	100 Main St.	Harrison
336-66-9999	Lindsay	175 Park Ave.	Pittsfield
019-28-3746	Smith	72 North St.	Rye

(a) The customer table

account-number	balance
A-101	500
A-215	700
A-102	400
A-305	350
A-201	900
A-217	750
A-222	700

(b) The account table

customer-id	account-number
192-83-7465	A-101
192-83-7465	A-201
019-28-3746	A-215
677-89-9011	A-102
182-73-6091	A-305
321-12-3123	A-217
336-66-9999	A-222
019-28-3746	A-201

(c) The depositor table

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DATABASE LANGUAGES

- Data-Definition Language (DDL)
- Data-Manipulation Language (DML)

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DATA-DEFINITION LANGUAGE (DDL)

- Specification notation for defining the database schema

Example: **create table** *instructor* (
 ID **char**(5),
 name **varchar**(20),
 dept_name **varchar**(20),
 salary **numeric**(8,2))

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DATA DICTIONARY 数据字典

- DDL compiler generates a set of table templates stored in a **data dictionary** 数据字典
 - Data dictionary contains **metadata**
 - (i.e., data about data, 元数据)
 - The data dictionary consists of record types (tables) created in the database by systems generated command files
- In MySQL
 - Data dictionary in **information_schema**

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DATA DICTIONARY IN MYSQL

The screenshot shows the MySQL Workbench interface. The 'Query' window contains the SQL statement '1 • show databases;'. The 'Result Grid' shows the output of the query, listing the databases: information_schema, mysql, performance_schema, saless, schema_test, sys, and world. The 'Schemas' pane on the left shows the same list of databases. The 'Output' pane at the bottom shows the execution details of the query.

DATA DICTIONARY IN MYSQL

The screenshot shows the MySQL Workbench interface. The 'Query' window contains a complex SQL query to retrieve metadata from the information_schema database. The query is as follows:

```
1 • SELECT a.TABLE_NAME "表",  
2 a.COLUMN_NAME "列", a.COLUMN_TYPE "类型",  
3 a.COLUMN_DEFAULT "默认值", a.IS_NULLABLE "是否为空",  
4 a.CHARACTER_SET_NAME "表字符集", a.COLLATION_NAME "校验字符集",  
5 CONCAT(a.COLUMN_COMMENT, a.COLUMN_KEY, a.EXTRA) "列备注",  
6 b.TABLE_COMMENT "表备注", b.ENGINE "引擎"  
7 FROM information_schema.COLUMNS a, information_schema.TABLES b  
8 WHERE a.TABLE_SCHEMA=b.TABLE_SCHEMA  
9 AND a.TABLE_SCHEMA='world' AND a.TABLE_NAME=b.TABLE_NAME;
```

The 'Result Grid' shows the output of the query, listing the tables and their columns with their respective metadata. The 'Schemas' pane on the left shows the same list of databases. The 'Output' pane at the bottom shows the execution details of the query.

DATA DICTIONARY IN MYSQL

- show databases;
- show tables;
- show columns from user;
- desc user;
- show indexes from user;
- select database();
- show privileges [for user];

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DATA-DEFINITION LANGUAGE (DDL)

- Data dictionary
 - Database schema
 - Consistency constraint
- Consistency constraint 一致性约束
 - Domain constraint 域约束
 - Referential Integrity 参照完整性
 - Primary key, foreign key
- Assertion 断言
- Authorization 授权: read, insert, update, delete, ...

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DATA-MANIPULATION LANGUAGE (DML)

- Language for accessing and manipulating the data organized by the appropriate data model
 - DML also known as **query language**

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DATA-MANIPULATION LANGUAGE (DML)

- Two classes of languages
 - **Pure** – used for proving properties about computational power and for optimization
 - 1. Relational Algebra
 - 2. Tuple元组 relational calculus演算
 - 3. Domain域 relational calculus
 - **Commercial** – used in commercial systems
 - SQL is the most widely used commercial language

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DATA-MANIPULATION LANGUAGE (DML)

- Types of DB Access
 - Retrieval of information stored in the database
 - Insertion of new information into the database
 - Deletion of information from the database
 - Modification of information stored in the database

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DATA-MANIPULATION LANGUAGE (DML)

- Procedural DMLs(过程式)
 - Require a user to specify what data are needed and how to get those data
- Declarative DMLs(声明式)
 - Nonprocedural DMLs
 - Require a user to specify what data are needed without specifying how to get those data

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RELATIONAL DATABASE

- Table
- Data-Manipulation Language
- Data-Definition Language
- Database Access from Application Programs

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RELATIONAL DATABASE

- TABLE

Columns			
ID	name	dept_name	salary
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

Rows

(a) The instructor table

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RELATIONAL DATABASE - TABLE

ID	name	dept_name	salary
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

(a) The instructor table

dept_name	building	budget
Comp. Sci.	Taylor	100000
Biology	Watson	90000
Elec. Eng.	Taylor	85000
Music	Packard	80000
Finance	Painter	120000
History	Painter	50000
Physics	Watson	70000

(b) The department table

RELATIONAL DATABASE

- DML

• SQL query language

- Structured Query Language

- Nonprocedural

• Example

Select instructor.name

From instructor

Where

instructor.dept_name = 'History';

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ID	name	dept_name	salary
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
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(a) The instructor table

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Comp. Sci.	Taylor	100000
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Elec. Eng.	Taylor	85000
Music	Packard	80000
Finance	Painter	120000
History	Painter	50000
Physics	Watson	70000

(b) The department table

RELATIONAL DATABASE - DML

Select instructor.ID,

department.dept_name

From instructor, department

Where

Instructor.dept_name =

department.dept_name

And

department.budget > 95000;

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ID	name	dept_name	salary
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
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(a) The instructor table

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Comp. Sci.	Taylor	100000
Biology	Watson	90000
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Music	Packard	80000
Finance	Painter	120000
History	Painter	50000
Physics	Watson	70000

(b) The department table

RELATIONAL DATABASE

- DDL

create table department(
dept_name char(20),
building char(15),
budget numeric(12,2));

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dept_name	building	budget
Comp. Sci.	Taylor	100000
Biology	Watson	90000
Elec. Eng.	Taylor	85000
Music	Packard	80000
Finance	Painter	120000
History	Painter	50000
Physics	Watson	70000

(b) The department table

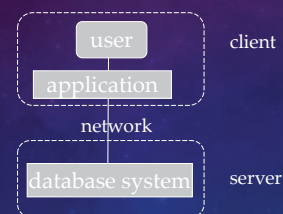
SQL STANDARD: ISO

<https://en.wikipedia.org/wiki/SQL>

Year	Name	Alias	Comments
1986	SQL-86	SQL-87	First formalized by ANSI.
1989	SQL-89	FIPS 127-1	Minor revision that added integrity constraints, adopted as FIPS 127-1.
1992	SQL-92	SQL2, FIPS 127-2	Major revision (ISO 9075). Entry Level SQL-92 adopted as FIPS 127-2.
1999	SQL 1999	SQL3	Added regular expression matching, recursive queries (e.g. transitive closure), triggers, support for procedural and control-of-flow statements, non-scalar types (arrays), and some object-oriented features (e.g. structured types). Support for embedding SQL in Java (SQL/JDBC) and vice versa (SQL/JRT).
2003	SQL 2003		Introduced XML-related features (SQL/XML), window functions, standardized sequences, and columns with auto-generated values (including identity-columns).
2006	SQL 2006		ISO/IEC 9075-14:2006 defines ways that SQL can be used with XML. It defines ways of importing and storing XML data in an SQL database, manipulating it within the database, and publishing both XML and conventional SQL data in XML form. In addition, it lets applications integrate queries into their SQL code with XQuery, the XML Query Language published by the World Wide Web Consortium (W3C), to concurrently access ordinary SQL data and XML documents [1].
2008	SQL 2008		Legalizes ORDER BY outside cursor definitions. Adds INSTEAD OF triggers. TRUNCATE statement [10] FETCH clause.
2011	SQL 2011		Adds temporal data (PERIOD FOR [12]) (more information at: Temporal database#History). Enhancements for window functions and FETCH clause [16].
2016	SQL 2016		Adds row pattern matching, polymorphic table functions, JSON.

RELATIONAL DATABASE

- DATABASE ACCESS FROM APPLICATION PROGRAMS



(a) Two-tier architecture

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RELATIONAL DATABASE

- DATABASE ACCESS FROM APPLICATION PROGRAMS

- Application Programs: C, C++, Java, Php, ...
- Ways to access db
 - By providing an application program interface (set of procedures) that can be used to send DML and DDL statements to the db and retrieve the results
 - ODBC (Open Database Connectivity), JDBC
 - By extending the host(宿主) language syntax to embed DML calls within the host language program
 - DML precompiler 预编译器

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DATABASE DESIGN

- Database design
 - Involves the design of the database schema
- Database design process
 - A high-level data model
 - conceptual framework: requirement
 - Choose a data model
 - logical-design, physical-design

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DATABASE DESIGN

- A HIGH-LEVEL DATA MODEL

What is an elephant?



盲人摸象
The Blind Men and
the Elephant

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DATABASE DESIGN: CASE

- A university organization
 - The **university** is organized into departments. Each department is identified by a unique name (dept name), is located in a particular building, and has a budget.
 - Each **department** has a list of courses it offers. Each course has associated with it a course id, title, dept name, and credits, and may also have associated prerequisites.
 - **Instructors** are identified by their unique ID. Each instructor has name, associated department (dept name), and salary.
 - **Students** are identified by their unique ID. Each student has a name, an associated major department (dept name), and tot cred (total credit hours the student earned thus far).
 - The university maintains a list of **classrooms**, specifying the name of the **building**, room number, and room capacity.

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DATABASE DESIGN: CASE

- A university organization (cont.,)
 - The university maintains a list of all **classes** (sections) taught. Each section is identified by a course id, sec id, year, and semester, and has associated with it a semester, year, building, room number, and time slot id (the time slot when the class meets).
 - The department has a list of teaching assignments specifying, for each instructor, the sections the instructor is teaching.
 - The university has a list of all student course registrations, specifying, for each student, the courses and the associated sections that the student has taken (registered for).

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DATABASE DESIGN: CASE

• E-R Model

- Entity
 - attribute
- Entity set
- Relationship
- Constraint
 - Mapping cardinalities 映射基数



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DATABASE DESIGN: CASE

- Normalization 规范化

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DATABASE DESIGN: CASE

- Normalization
- Is there any problem with this relation?

ID	name	salary	dept_name	building	budget
22222	Einstein	95000	Physics	Watson	70000
12121	Wu	90000	Finance	Painter	120000
32343	El Said	60000	History	Painter	50000
45565	Katz	75000	Comp. Sci.	Taylor	100000
98345	Kim	80000	Elec. Eng.	Taylor	85000
76766	Crick	72000	Biology	Watson	90000
10101	Srinivasan	65000	Comp. Sci.	Taylor	100000
58583	Califieri	62000	History	Painter	50000
83821	Brandt	92000	Comp. Sci.	Taylor	100000
15151	Mozart	40000	Music	Packard	80000
33456	Gold	87000	Physics	Watson	70000
76543	Singh	80000	Finance	Painter	120000

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DATABASE DESIGN: CASE

- Normalization theory
 - Functional dependency 函数依赖
 - e.g. $dept_name \rightarrow building, budget$

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DATABASE DESIGN

- Data Storage and Querying
 - Storage manager
 - Storage space
 - Query processor
 - Performance

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DATABASE DESIGN - STORAGE MANAGER

- Storage Manager
 - Authorization and integrity manager
 - Transaction manager
 - File manager
 - Buffer manager

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DATABASE DESIGN - STORAGE MANAGER

- Storage manager
 - a program module that provides the interface between the low-level data stored in the database and the application programs and queries submitted to the system.
 - The storage manager is responsible to the following tasks:
 - Interaction with the OS file manager
 - Efficient storing, retrieving and updating of data

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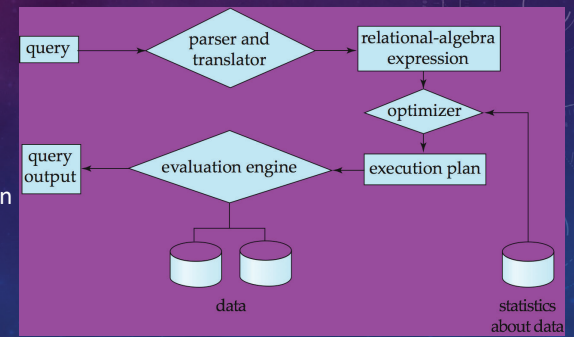
DATABASE DESIGN - STORAGE MANAGER

- Physical implementation of DB Storage
 - Data files, which store the database itself
 - Data dictionary
 - Indices(索引)

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DATABASE DESIGN- QUERY PROCESSOR

- Parsing and translation
- Optimization
- Evaluation



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DATABASE DESIGN - QUERY PROCESSOR

- Query Processor
 - DDL interpreter
 - DML interpreter
 - A query can be translated into any of a number of alternative evaluation plans
 - Query optimization
 - Query evaluation engine 查询执行引擎

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DATABASE DESIGN - TRANSACTION MANAGEMENT

- Transaction 事务
 - A collection of operations that performs a single logical function in a database application
 - Example:
 - transfer funds from A to B
 - User A buys a book from user B

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DATABASE DESIGN - TRANSACTION MANAGEMENT

- The ACID Properties of Transactions
 - Atomicity 原子性
 - Consistency 一致性
 - Isolation 独立性
 - Each transaction must appear to be executed as if no other transaction is executing at the same time
 - Durability 持久性

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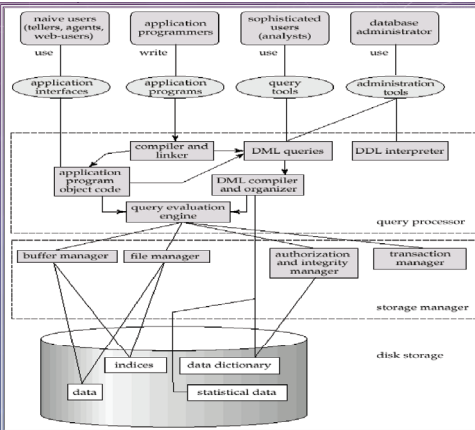
DATABASE DESIGN - TRANSACTION MANAGEMENT

- Transaction Management
 - Logging 记日志
 - Concurrency control
 - Deadlock resolution
 - Recovery manager
 - Failure recovery

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DATABASE ARCHITECTURE

• Components of a DB



DATABASE USERS AND ADMINISTRATORS

• Database Users and User Interfaces

- Naïve users
- Application programmers
- Sophisticated users
- Specialized users

DATABASE USERS AND ADMINISTRATORS

• Database Administrator 数据库管理员

- Schema definition
- Storage structure and access-method definition
- Schema and physical-organization modification
- Granting of authorization for data access
- Routine maintenance
- Backup, Enough free disk space, Monitoring jobs

DATABASE APPLICATION ARCHITECTURE

- The architecture of a database systems is greatly influenced by the underlying computer system on which the database is running:
 - Centralized 集中式
 - Client-server 客户服务器式
 - Parallel (multi-processor) 并行: multi-processors
 - Distributed 分布式

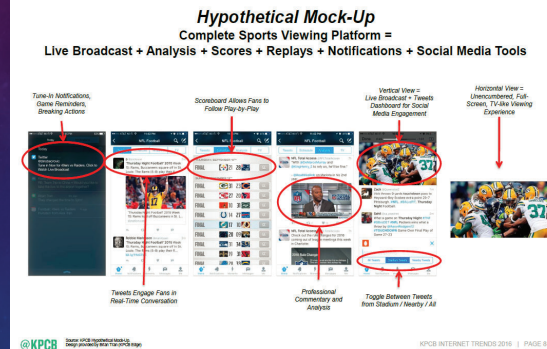
INFORMATION RETRIEVAL

• Information Retrieval 信息检索

- Querying of unstructured textual data
- Textual data is unstructured, unlike the rigidly structured data in relational databases
- Textual data has grown explosively

INFORMATION RETRIEVAL: TODAY

2016E = Milestone Year for 'Traditional' Live Streaming on Social Networks...
NFL Live Broadcast TV of Thursday Night Football on Twitter (Fall 2016)



DATA WAREHOUSE

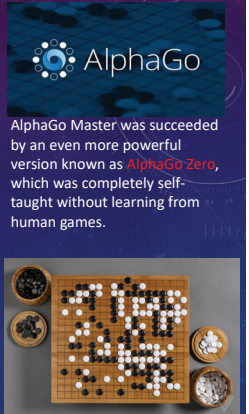
- Data warehouse
 - 数据仓库
 - Gather data from multiple sources under a unified schema, at a single site
 - Then, they provide the user a single uniform interface to data

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DATA MINING

- Refers loosely to the process of semi-automatically analyzing large DB to find useful patterns
 - Case: beer and nappy
- i.e. **Knowledge discovery** in artificial intelligence: machine learning
- 聚类分析CA、决策树ID、神经网络算法NN、贝叶斯网络方法BN、支持向量机SVG、...

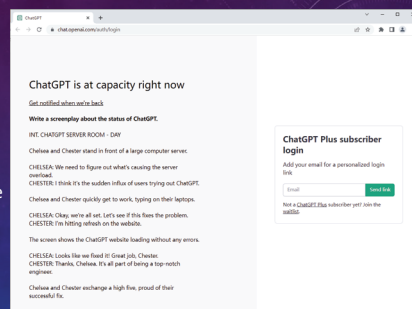
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AI : DATA-DRIVEN

- Artificial General intelligence
 - AI is intelligence—perceiving, synthesizing, and inferring information—demonstrated by machines, as opposed to intelligence displayed by non-human animals and humans.
 - Example tasks in which this is done include speech recognition, computer vision, translation between (natural) languages, as well as other mappings of inputs.
 - ChatGPT, DeepSeek, ...

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ChatGPT is, in essence, a simple online artificial intelligence chatbot created by OpenAI in December 2022. You can ask it a question, and it will answer that question. Improving Language Understanding by **Generative Pre-Training**



HISTORY OF DATABASE SYSTEMS

- 1950s and early 1960s: tape
- Late 1960s and 1970s
 - Hard disk, file, DB(hierarchical, network)
 - network db
 - CODASYL, Integrated Data Store (IDS)
- Codd, E. F.. "A relational model of data for large shared data banks." Communications of The ACM 13.6 (1970): 377-387.

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HISTORY OF DATABASE SYSTEMS

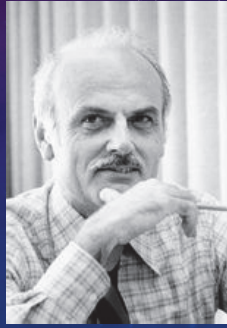
- 1980s
 - System R: IBM
 - Astrahan, Morton M., et al. "System R: relational approach to database management." ACM Transactions on Database Systems 1.2 (1976): 97-137.
 - Ingres: BSD
 - IBM DB2, Oracle, DEC Rdb
- Early 1990s
 - Object-relational DB

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EDGAR F. CODD

• Edgar Frank Codd

- Edgar Frank "Ted" Codd was an English computer scientist who, while working for IBM, invented the relational model for database management, the theoretical basis for relational databases
- ACM Turing Award, 1981
- Online analytical processing (OLAP)
- 19 August 1923 – 18 April 2003



HISTORY OF DATABASE SYSTEMS

• 1990s

- WWW: DB had to support Web interface to data

• 2000s

- XML, Xquery
- PostgreSQL, MySQL
- Giant data storage systems
 - Google BigTable, Yahoo PNuts, Amazon, ...

<https://db-engines.com/en/ranking>

RANK OF DATABASES

424 systems in ranking, February 2025

Rank			DBMS	Database Model	Score		
Feb 2025	Jan 2025	Feb 2024			Feb 2025	Jan 2025	Feb 2024
1.	1.	1.	Oracle	Relational, Multi-model	1254.82	-3.93	+13.38
2.	2.	2.	MySQL	Relational, Multi-model	999.99	+1.84	-106.67
3.	3.	3.	Microsoft SQL Server	Relational, Multi-model	786.87	-11.69	-66.70
4.	4.	4.	PostgreSQL	Relational, Multi-model	659.62	-3.79	+30.21
5.	5.	5.	MongoDB	Document, Multi-model	396.63	-5.87	-23.73
6.	7.	6.	Redis	Key-value, Multi-model	157.91	+4.55	-2.80
7.	6.	9.	Snowflake	Relational	155.58	+1.68	+28.13
8.	8.	7.	Elasticsearch	Multi-model	134.63	-0.29	-1.11
9.	9.	8.	IBM Db2	Relational, Multi-model	125.44	+2.46	-6.79
10.	10.	10.	SQLite	Relational	113.82	+7.13	-3.47
11.	11.	12.	Apache Cassandra	Wide column, Multi-model	102.58	+3.39	-6.69

<https://www.modb.pro/dbRank>

RANK OF DATABASES IN CHINA

排名	上月	半年前	数据库	模型	数据库类型	部署方式	商业模式	年份	初次	案例	用户	评价	得分
1	1	1	Oracle	关系型	商业	商业	商业	151	26	26	9	1	802.24
2	2	1	PolarDB	关系型	商业	商业	商业	582	70	10	10	2	777.41
3	3	2	openGauss	关系型	商业	商业	商业	573	11	16	8	5	660.24
4	4	3	TiDB	关系型	商业	商业	商业	40	54	18	6	1	584.11
5	5	4	人大金仓	关系型	商业	商业	商业	333	8	12	6	3	582.77
6	7	6	达梦数据库	关系型	商业	商业	商业	318	8	8	11	0	577.77
7	7	7	GAUSS	关系型	商业	商业	商业	181	1	46	9	9	524.77
8	8	8	Geared	关系型	商业	商业	商业	830	14	9	7	4	434.24
9	9	9	TDSQL	关系型	商业	商业	商业	89	18	12	9	8	304.52
10	10	10	GoldenDB	关系型	商业	商业	商业	181	28	28	7	1	300.24

SUMMARY

VLDB、SIGMOD、ICDE

- Database system
- Data Abstraction
- Database Model
- Database Languages
- Relational Database
- Database Design
- Database Architecture
- History of Database

Very
Large
Data
Bases



Q&A?

赞数字文明

李旭东

文明初叶几时真，造化阴阳始幻尘。
书简成山薪火旺，零壹遁迹智能春。
有形百载多将朽，数字千年总是新。
懵懂蹒跚别旧日，一朝奋起笑前津。



THANKS!

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